

# Mechanics

## Kinematics (Motion in a Straight Line)

### Key quantities:

- $s$  = displacement (m)
- $u$  = initial velocity (m/s)
- $v$  = final velocity (m/s)
- $a$  = acceleration ( $\text{m/s}^2$ )
- $t$  = time (s)

## SUVAT Equations (Constant Acceleration)

| Equation                   | Missing Variable |
|----------------------------|------------------|
| $v = u + at$               | $s$              |
| $s = ut + \frac{1}{2}at^2$ | $v$              |
| $s = vt - \frac{1}{2}at^2$ | $u$              |
| $v^2 = u^2 + 2as$          | $t$              |
| $s = \frac{1}{2}(u + v)t$  | $a$              |

### Worked Example

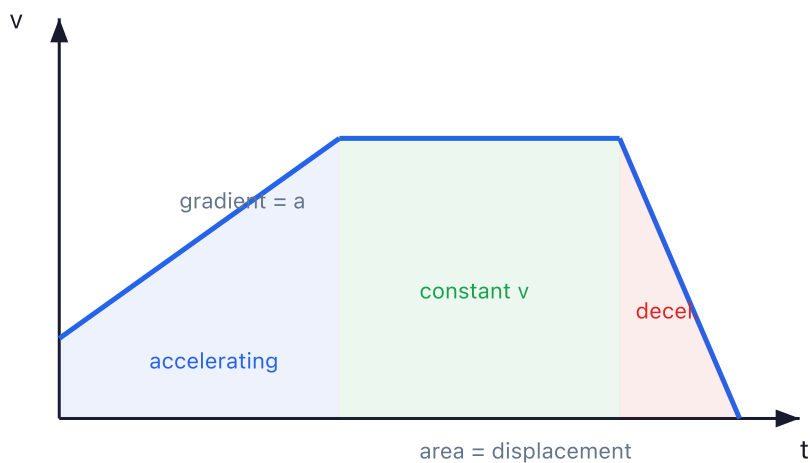
A particle starts from rest and accelerates at  $10 \text{ m/s}^2$  for 4 s. Find the distance travelled.

$u = 0, a = 10, t = 4$ , find  $s$ :

$$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(10)(16) = 80 \text{ m}$$

## Velocity-Time Graphs

- **Gradient** = acceleration
- **Area under the graph** = displacement



**Exam Tip:** SUVAT only works for **constant acceleration**. If acceleration varies, use calculus:  $v = \frac{ds}{dt}$ ,  $a = \frac{dv}{dt}$ ,  $s = \int v dt$ .

## Variable Acceleration

When acceleration is not constant, use calculus:

$$v = \frac{ds}{dt} \quad a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

$$s = \int v dt \quad v = \int a dt$$

### Worked Example: Differentiation

A particle has velocity  $\mathbf{v} = (3t^2)\mathbf{i} + (2t)\mathbf{j}$ . Find the acceleration at  $t = 2$ .

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (6t)\mathbf{i} + 2\mathbf{j}$$

$$\text{At } t = 2 : \quad \mathbf{a} = 12\mathbf{i} + 2\mathbf{j}$$

### Worked Example: Integration to Find Displacement

A particle has acceleration  $a = (6t - 2) \text{ m/s}^2$ . At  $t = 0$ , the velocity is 3 m/s and displacement is 0. Find the displacement when  $t = 3$ .

#### Step 1: Find velocity

$$v = \int (6t - 2) dt = 3t^2 - 2t + c$$

At  $t = 0$ ,  $v = 3$ :  $c = 3$ . So  $v = 3t^2 - 2t + 3$ .

#### Step 2: Find displacement

$$s = \int (3t^2 - 2t + 3) dt = t^3 - t^2 + 3t + k$$

At  $t = 0$ ,  $s = 0$ :  $k = 0$ . At  $t = 3$ :

$$s = 27 - 9 + 9 = 27 \text{ m}$$

## Forces and Newton's Laws

### Newton's Three Laws

1. An object remains at rest or in uniform motion unless acted on by a resultant force
2.  $\mathbf{F} = m\mathbf{a}$  — the resultant force equals mass times acceleration
3. Every action has an equal and opposite reaction

### Common Forces

| Force           | Symbol      | Direction                          |
|-----------------|-------------|------------------------------------|
| Weight          | $W = mg$    | Vertically downward                |
| Normal reaction | $R$ or $N$  | Perpendicular to surface           |
| Friction        | $F = \mu R$ | Opposite to motion/tendency        |
| Tension         | $T$         | Along the string, away from object |

### Worked Example

A 5 kg block on a rough surface ( $\mu = 0.4$ ) is pulled with 30 N horizontally. Find the acceleration.

Normal reaction:  $R = mg = 5 \times 9.8 = 49 \text{ N}$

Friction:  $F = \mu R = 0.4 \times 49 = 19.6 \text{ N}$

$$a = \frac{30 - 19.6}{5} = \frac{10.4}{5} = 2.08 \text{ m/s}^2$$

## Connected Particles

### Pulleys

For two particles of mass  $m_1$  and  $m_2$  connected by a string over a smooth pulley (with  $m_1 > m_2$ ):

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} \quad T = \frac{2m_1m_2g}{m_1 + m_2}$$

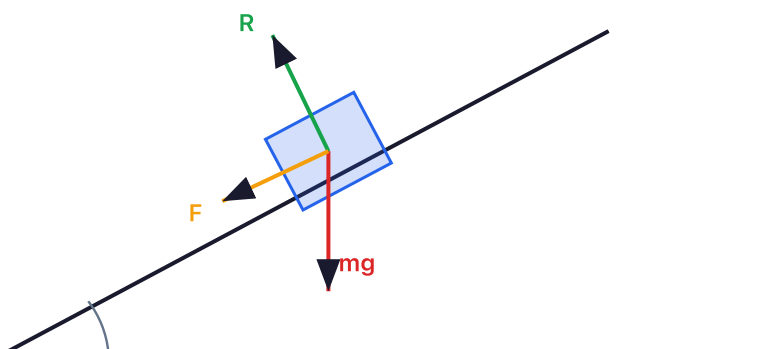
### Worked Example

Masses of 3 kg and 2 kg are connected over a smooth pulley. Find the acceleration ( $g = 10$ ).

$$a = \frac{(3 - 2) \times 10}{3 + 2} = \frac{10}{5} = 2 \text{ m/s}^2$$

**Exam Tip:** For connected particle problems, draw a **separate force diagram** for each particle and apply  $F = ma$  to each. The string tension is the same throughout (for a light, inextensible string).

## Motion on Inclined Planes



Resolve forces parallel and perpendicular to the slope:

- **Along the plane:**  $mg \sin \alpha$  (down the slope)
- **Perpendicular:** Normal reaction  $R = mg \cos \alpha$
- **Friction:**  $F = \mu R = \mu mg \cos \alpha$

### Worked Example

A 3 kg object slides down a smooth incline at  $30^\circ$ . Find its acceleration.

$$a = g \sin 30^\circ = 9.8 \times 0.5 = 4.9 \text{ m/s}^2$$

## Projectiles

A projectile launched at speed  $u$  at angle  $\theta$  to the horizontal:

- **Horizontal:** constant velocity  $u \cos \theta$  (no horizontal acceleration)
- **Vertical:** initial velocity  $u \sin \theta$ , acceleration  $-g$

## Key Formulae

$$\text{Time of flight} = \frac{2u \sin \theta}{g}$$

$$\text{Maximum height} = \frac{u^2 \sin^2 \theta}{2g}$$

$$\text{Range} = \frac{u^2 \sin 2\theta}{g}$$

### Worked Example: Full Projectile Problem

A ball is projected from ground level at  $30^\circ$  to the horizontal with speed 20 m/s. Find: (a) the time of flight, (b) the maximum height, (c) the range. ( $g = 10 \text{ m/s}^2$ )

**Horizontal:**  $v_x = 20 \cos 30^\circ = 10\sqrt{3} \text{ m/s}$  (constant)

**Vertical:**  $u_y = 20 \sin 30^\circ = 10 \text{ m/s}$ ,  $a = -10 \text{ m/s}^2$

**(a) Time of flight:** Land when  $s_y = 0$ :

$$s_y = u_y t + \frac{1}{2} a t^2 = 10t - 5t^2 = 0 \implies t(10 - 5t) = 0 \implies t = 2 \text{ s}$$

**(b) Maximum height:** At top,  $v_y = 0$ :

$$v_y^2 = u_y^2 + 2as \implies 0 = 100 - 20H \implies H = 5 \text{ m}$$

**(c) Range:**

$$R = v_x \times t = 10\sqrt{3} \times 2 = 20\sqrt{3} \approx 34.6 \text{ m}$$

**Exam Tip:** Treat horizontal and vertical motion separately. The horizontal velocity is constant (no air resistance). Use SUVAT vertically with  $a = -g$  (taking upward as positive).

## Moments

**Moment of a force** about a point = force  $\times$  perpendicular distance from the point to the line of action.

$$M = F \times d$$

For equilibrium:

- Resultant force = 0
- Sum of clockwise moments = sum of anticlockwise moments (about any point)

### Worked Example

A uniform beam of length 6 m and weight 120 N rests on supports at its ends A and B. A 60 N weight is placed 2 m from A. Find the reaction at A.

Taking moments about B:

$$R_A \times 6 = 120 \times 3 + 60 \times 4 = 360 + 240 = 600$$

$$R_A = 100 \text{ N}$$

Check:  $R_B = 120 + 60 - 100 = 80 \text{ N}$ .

## Momentum and Impulse [Further Maths only — not on the AQA A-level Maths spec]

**Momentum:**  $p = mv$  (vector quantity, units: kg m/s or N s)

**Impulse:**  $I = Ft = mv - mu$  (change in momentum)

**Conservation of momentum:** In a closed system, total momentum before = total momentum after.

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

## Coefficient of Restitution

$$e = \frac{\text{speed of separation}}{\text{speed of approach}} \quad 0 \leq e \leq 1$$

$e = 1$ : perfectly elastic.  $e = 0$ : perfectly inelastic (objects coalesce).

### Worked Example

A 0.5 kg ball moving at 4 m/s strikes a wall and rebounds at 3 m/s. Find the impulse.

Taking rebound direction as positive:

$$I = mv - mu = 0.5(3) - 0.5(-4) = 1.5 + 2 = 3.5 \text{ N s}$$

## Work, Energy, and Power [Further Maths only — not on the AQA A-level Maths spec]

**Work done:**  $W = Fs \cos \theta$  (force  $\times$  displacement  $\times$  cosine of angle between them)

**Kinetic energy:**  $KE = \frac{1}{2}mv^2$

**Gravitational PE:**  $GPE = mgh$

**Power:**  $P = \frac{W}{t} = Fv$

## Work-Energy Principle

Net work done on a body = change in kinetic energy:

$$W_{\text{net}} = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

## Conservation of Energy

Total energy is conserved (KE + PE + work done against friction):

$$\frac{1}{2}mu^2 + mgh_1 = \frac{1}{2}mv^2 + mgh_2 + F_{\text{friction}} \times d$$

## Worked Example

A 2 kg block slides 5 m down a rough incline ( $30^\circ$ ,  $\mu = 0.3$ ). Find its speed at the bottom (starting from rest).

Height lost:  $h = 5 \sin 30^\circ = 2.5 \text{ m}$

Friction:  $F = \mu mg \cos 30^\circ = 0.3 \times 2 \times 9.8 \times \cos 30^\circ \approx 5.09 \text{ N}$

$$\frac{1}{2}(2)v^2 = 2(9.8)(2.5) - 5.09(5)$$

$$v^2 = 49 - 25.45 = 23.55 \implies v \approx 4.85 \text{ m/s}$$

**Exam Tip:** In energy problems, always draw a clear diagram and define your reference level for GPE. The work-energy approach often avoids the need to find acceleration — useful when the force varies or the path is complex.

## Resolving Forces

Any force can be resolved into two perpendicular components:

$$F_x = F \cos \theta \quad F_y = F \sin \theta$$

The resultant of two perpendicular forces  $F_x$  and  $F_y$ :

$$R = \sqrt{F_x^2 + F_y^2} \quad \tan \alpha = \frac{F_y}{F_x}$$

### Worked Example

Two forces of 5 N and 12 N act perpendicular to each other. Find the resultant.

$$R = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13 \text{ N}$$